Tokenisation of Residential Rental Income Using Fungible Tokens: A Decentralised Finance Approach

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1. Introduction

This report designs a tokenisation scheme for residential rental income using a fungible token structure. The proposal starts from the economics of the underlying asset and explains how the asset is abstracted into a token, how the token functions in a market environment, and what risks arise from the design. The aim is not simply to digitise ownership, but to create a financially meaningful instrument that improves access, transferability and potential liquidity while preserving a clear relationship to the underlying asset.

Residential real estate is suitable for tokenisation because it combines stable cash-flow characteristics with structural market inefficiencies (Catalini and Gans, 2020). Rental property can generate regular income and may increase in value over time due to market demand dynamics (Geltner et al., 2013). At the same time, traditional property investment remains difficult for many participants due to high minimum capital requirements, slow transaction processes, and limited secondary market liquidity. Tokenisation offers a possible solution by dividing economic rights into smaller digital units that can be issued, transferred and recorded on blockchain infrastructure (McMahan, 2006).

The proposed design uses fungible tokens because the rights attached to each unit are economically identical. Each token holder receives a proportional claim on a pool of rental income and, where applicable, a proportional claim on sale proceeds after costs. The model is therefore closer to a fractional income share instrument than to a unique collectable asset.

2. Asset Description and Value Source

The selected asset is a portfolio of residential properties that generate rental income. The main source of value is the cash flow received from tenants after deducting operating expenses such as maintenance, insurance, management fees and other property based costs (Geltner et al., 2013). In addition to income, the asset may create value through long term capital appreciation if the properties' market values rise over time. The tokenisation design in this report focuses primarily on the income stream, as recurring cash flow provides a clearer basis for token holders' rights and valuation than speculative appreciation alone.
From an economic perspective, rental property produces value in three ways.

First, it offers a recurring yield through rent payments. This makes it attractive to investors who prefer periodic returns rather than relying only on future resale gains (McMahan, 2006).

Second, the underlying real estate can preserve or increase value over the longer term if demand for housing remains strong relative to supply (Geltner et al., 2013).

Third, residential property can provide some diversification benefits because its performance drivers differ from those of many purely digital or short duration financial assets.

However, the asset also has weaknesses that make tokenisation attractive. Real estate is indivisible in its conventional form, meaning investors usually need substantial capital to gain exposure. Transfer processes are also slow because sales involve lawyers, brokers, due diligence and settlement procedures. The market is therefore less accessible and less liquid than many conventional securities. These characteristics support the case for transforming economic claims on rental income into smaller and more easily transferable digital units.

3. Rationale for Tokenisation

The rationale for tokenising residential rental income is based on four linked objectives: improving access, increasing transferability, supporting liquidity and enhancing transparency.

First, tokenisation allows economic rights to be divided into many smaller units. Instead of buying an entire property, an investor can buy a limited number of tokens representing only a small proportion of the income stream (Catalini and Gans, 2020). This lowers the barrier to entry and widens the potential investor base.

Second, tokenisation improves transferability. In a traditional property investment, an investor cannot easily sell a small fraction of ownership without a bespoke legal arrangement. A fungible token design standardises those rights into interchangeable units that can be moved between wallets and traded on supported platforms. Standardisation is particularly important because liquidity usually emerges more readily when claims are simple, uniform and comparable (Nakamoto, 2008).

Third, tokenisation may improve liquidity by making trading easier and cheaper than direct property transactions (Tapscott and Tapscott, 2016). A residential building might take weeks or months to sell, but tokens representing proportional claims could be traded more quickly in smaller quantities. This does not guarantee deep liquidity, but it can reduce friction and support more frequent price discovery.

Fourth, blockchain infrastructure can improve transparency. Transfers, balances and issuance rules can be recorded on chain, giving participants visibility over supply and transaction history (Nakamoto, 2008). Smart contracts can also automate administrative tasks, such as income distribution calculations, reducing manual processing and lowering operational risk in some areas.

The target holders for this instrument include small investors seeking property exposure, income focused investors seeking periodic distributions, and globally distributed users seeking a more accessible route to real estate based returns. The token is therefore designed as a financial access product rather than as a consumption, membership or collectable asset.

4. Token Structure and Design

A fungible token structure is the most suitable choice because each token carries the same economic rights. Every unit represents an equal fractional entitlement to a defined rental income pool and, if the asset is sold, to net sale proceeds according to the holder's proportional share. An NFT structure would be less appropriate, as NFTs are better suited to unique assets or differentiated rights, whereas this model requires standardised, divisible claims.
The proposed token can be implemented using an ERC-20-style smart contract, which reflects standardised token design principles in blockchain systems (Nakamoto, 2008). A fixed supply is minted at issuance, with the total number of tokens corresponding to the value of the underlying portfolio or to a financing target set by the issuer. For example, if the portfolio is represented through one million tokens, each token corresponds to one millionth of the defined economic interest. Divisibility allows investors to hold very small positions, thereby supporting access and potentially increasing market breadth.

The rights embedded in the token should be clearly stated. The core right is a proportional claim on distributable rental income after approved deductions. A secondary right may be a proportional claim on net proceeds if the properties are refinanced or sold, although this should be governed by the legal wrapper around the tokenisation structure. The token itself does not create real world ownership rights; rather, it represents rights that must be recognised through associated legal documentation and governance arrangements.

Transferability should be enabled, with compliance controls in place if required. A whitelist model may be used if the issuer needs to restrict transfers to verified participants. This slightly reduces openness but can help align the scheme with legal requirements. Supply should generally remain fixed after initial issuance to avoid diluting holders without a clear governance process. A burn mechanism may be included if tokens are redeemed following asset liquidation.

5. Market Access and Liquidity Design

The market design begins with a primary issuance. In this stage, tokens are sold to investors in exchange for capital used to acquire, refinance, or support the underlying residential portfolio. The initial offering price can be derived from a valuation model that considers expected rental cash flow, property quality, vacancy assumptions, operating costs and any structural fees. Clear disclosure at issuance is essential because investors need to understand what the token represents and what it does not represent.

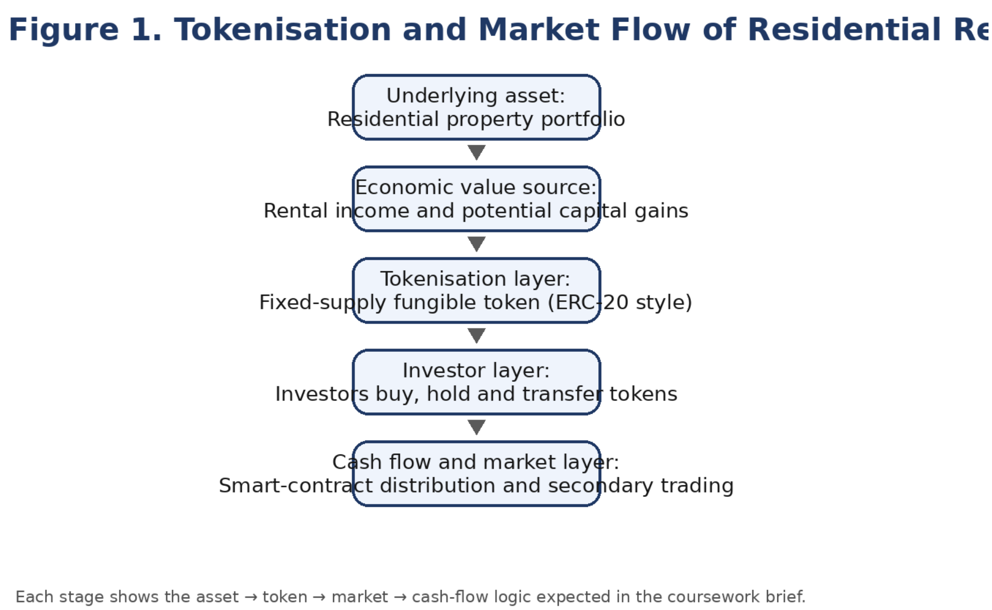
Once issued, the token may be admitted to a secondary trading venue such as a decentralised exchange, a permissioned marketplace or a specialised tokenisation platform. The most realistic route for an asset linked instrument may be a controlled platform rather than a fully open venue, because compliance, investor verification and disclosure obligations may apply. Even so, blockchain based transfer infrastructure can still support faster settlement and lower operational friction than conventional property transfers.

Liquidity in this model arises from divisibility, standardisation and lower transfer friction. Investors can trade smaller positions, which is much easier than selling part of a building. Market access may also broaden, as participants from different regions can interact with the tokenised instrument via digital infrastructure. In theory, this should support stronger price discovery and narrower holding constraints.

However, liquidity emerges from market participation (Catalini and Gans, 2020). Tokens linked to real-world assets often face a liquidity paradox: they are technically easy to trade, but economically hard to trade in size unless there is a sufficient base of active buyers and sellers. Liquidity can therefore improve relative to direct property investment, without reaching the depth of major public equity markets. This is an important limitation because marketing tokenisation as a complete solution to illiquidity would be misleading.

Pricing in secondary markets is also unlikely to depend only on property fundamentals. Token prices may reflect expected rental yields, interest rate conditions, perceived quality of management, legal confidence in the structure, and the broader sentiment of the digital asset market. The token may therefore trade at a premium or discount to the estimated net asset value of the underlying properties.

Figure 1. Tokenisation and Market Flow of Residential Rental Income



The diagram above summarises the asset → token → market → cash-flow chain.

6. Risk Analysis and Limitations

Risk analysis is central to this proposal because the success of tokenisation depends not only on technical feasibility but also on whether the design can withstand legal, market and operational stress. The first set of risks is technical. Smart contracts may contain coding errors, insecure logic or vulnerabilities that can be exploited (Szabo, 1997). Even if the core token contract is simple, surrounding infrastructure such as wallets, bridges, custody arrangements or admin controls may introduce attack surfaces.

The second category is asset and market risk. Rental income is not guaranteed. Vacancy, tenant default, maintenance shocks, regulatory changes in housing markets or declining local demand can reduce cash flow (Arner et al., 2017). Interest-rate increases may also reduce property valuations and lower investor demand for yield based products. Because token holders are economically exposed to these underlying factors, tokenisation does not remove the real risks of real estate; it mainly changes the format in which exposure is packaged and traded.

Third, there is regulatory and legal risk. A token that gives holders rights to income or proceeds may be treated as a security or a security like instrument depending on the jurisdiction. This creates compliance obligations relating to offering rules, investor eligibility, disclosures and trading restrictions. The legal wrapper around the token is therefore as important as the smart contract itself.

Fourth, there is a structural limitation arising from the link between on-chain and off-chain reality. Rental payments are generated in the physical world and must be reported to the blockchain system before they can be distributed. This creates an oracle problem: token holders rely on the accuracy and honesty of external information feeds or managers. The system therefore remains partially dependent on trusted intermediaries even though it uses decentralised infrastructure for record-keeping and transfer.

Fifth, there is a liquidity limitation. Tokenisation is often promoted as if technical tradability automatically creates genuine liquidity. In reality, a market can remain thin, with wide spreads and limited depth. Under stress, holders may all wish to sell at the same time while few buyers appear. This could cause prices to diverge significantly from the estimated property value. Finally, there are governance and agency risks. The manager responsible for property operations, rent collection and reporting still plays a central role. If incentives are weak or conflicts of interest are poorly controlled, token holders may bear the cost. Effective tokenization therefore, requires not only code but also sound governance, disclosure and accountability.

7. Role of DeFi and Blockchain Infrastructure

Blockchain infrastructure supports this tokenisation scheme by providing a shared ledger for issuance, balance tracking and transfers. Smart contracts can codify supply rules, automate certain administrative actions and reduce settlement friction. In practical terms, this means that token balances can update transparently, transfers can be recorded in near real time, and income-distribution logic can be executed according to predefined rules rather than through fully manual processes. (Nakamoto, 2008)

DeFi infrastructure can further enhance the model. A decentralised exchange or permissioned automated market maker could provide a venue for secondary trading. Lending protocols might allow token holders to post the asset as collateral, increasing capital efficiency for some users. On-chain settlement also reduces dependence on long conventional settlement chains (Tapscott and Tapscott, 2016). These features make the instrument more versatile than a static paper-based fractional property scheme.

At the same time, DeFi introduces its own risks. Automated market makers can be inefficient for less liquid real-world-asset tokens, collateralised lending can amplify leverage, and interoperability with broader crypto markets can transmit volatility to an asset that might otherwise appear relatively stable. For this reason, DeFi should be viewed as enabling infrastructure rather than as a guarantee of superior outcomes.

8. Smart Contract Implementation

The implementation can be represented through a simplified ERC-20 smart contract. The contract would define the token name, symbol, total supply and balances, while enabling transfers and approved third-party transfers. A separate income-distribution mechanism can record distributable amounts and allocate rata claims across token holders. For coursework purposes, the contract demonstrates how standardised blockchain logic can represent financial rights programmatically.

However, the smart contract should be described carefully. It does not replace the legal agreements linking the token to the underlying property portfolio. Instead, it operationalises the scheme's token layer (Szabo, 1997). The contract address and GitHub repository provide evidence of implementation, while screenshots or a testnet deployment show that the model has been executed in practice under the ***Appendix*** section below

8.1 Implementation Procedure

The smart contract was implemented and deployed using a structured development workflow involving MetaMask, the Sepolia Ethereum test network, and Remix IDE.

First, MetaMask was used as a browser based wallet to manage blockchain interactions and sign transactions. The wallet was configured to connect to the Sepolia test network, which provides a simulated Ethereum environment for testing smart contracts without using real funds.

Next, test Ether (SepoliaETH) was obtained from a faucet to enable transaction execution. This allowed the deployment of the smart contract without incurring real financial cost.

The contract was then developed, compiled, and deployed using Remix IDE, an online Solidity development environment. The “Browser Extension” environment was selected in Remix to connect the IDE with MetaMask. This enabled the contract to be deployed directly onto the Sepolia network.

During deployment, the constructor parameter (_initialSupply) was set to 1,000,000 tokens, representing the total supply of the fungible token. The deployment transaction was confirmed through MetaMask, and upon successful execution, a unique contract address was generated.

This process ensured that the smart contract was not only theoretically designed but also practically validated in a live blockchain test environment. (*Please see Screenshot 1-4 under the Appendices Section*)

9. Conclusion

This report has proposed a tokenisation design for residential rental income using fungible tokens. The choice of asset is justified by the combination of stable cash flow potential and structural market inefficiencies in traditional real-estate investing. A fungible token is the appropriate design because each holder receives identical proportional rights, making divisibility, transferability and secondary-market trading more feasible than under a non-fungible structure.

The proposal shows that tokenisation can improve access, standardise investment claims and support faster transfer and broader participation. It also demonstrates how blockchain and DeFi infrastructure can enhance settlement, transparency and programmability. Nevertheless, the model has important limitations. Legal enforceability remains crucial, off chain data dependencies weaken pure decentralisation, underlying property risks remain fully relevant, and liquidity may be shallower than promotional narratives suggest.

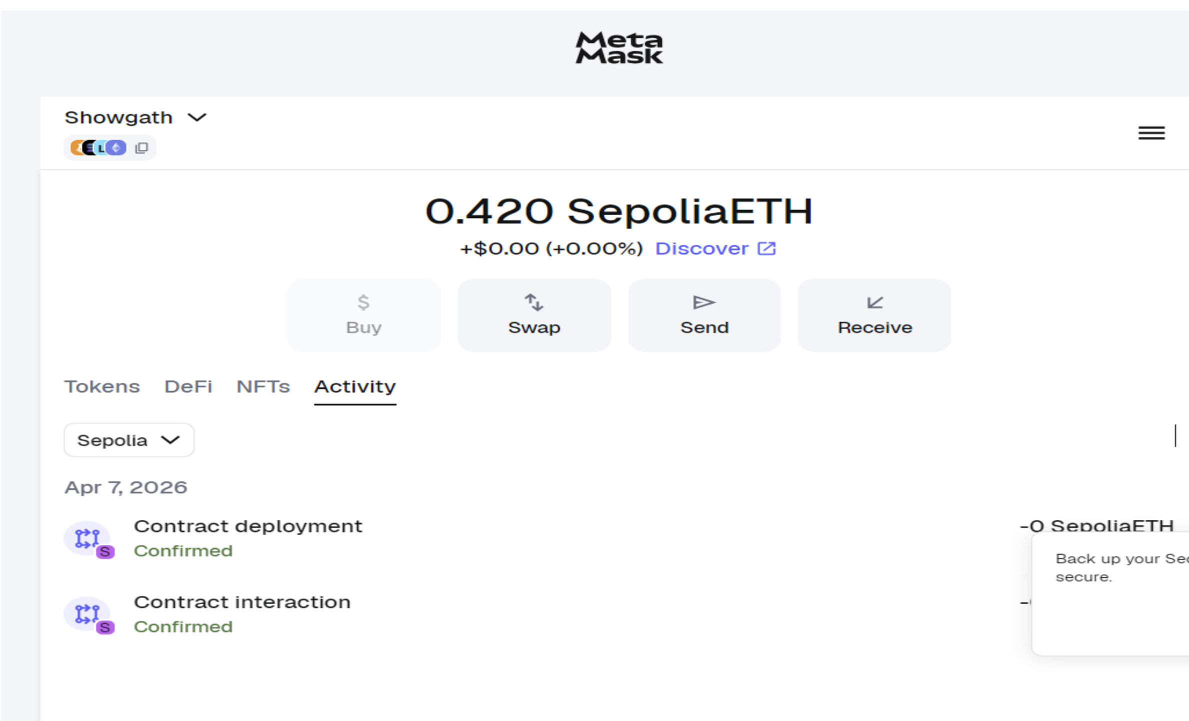
The strongest conclusion is therefore balanced rather than purely optimistic. Tokenisation is valuable only when the token design creates a coherent connection among asset economics, investor rights, market function, and risk management.

Appendix Placeholders

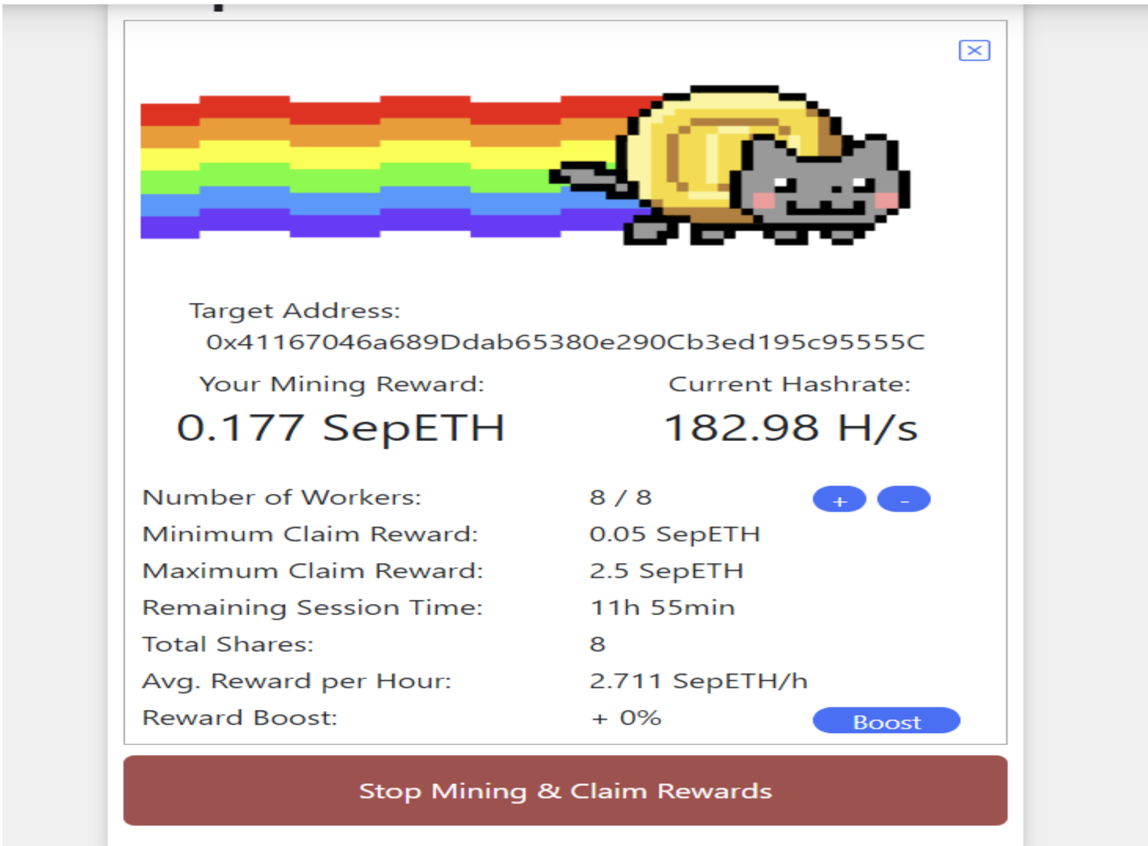
GitHub Repository: [https://github.com/Showgath/tokenisation-project.git]

Testnet Contract Address: [0x98405Ba1aC2FED495Fc1169fA2E23f3D4EEEF0AB]

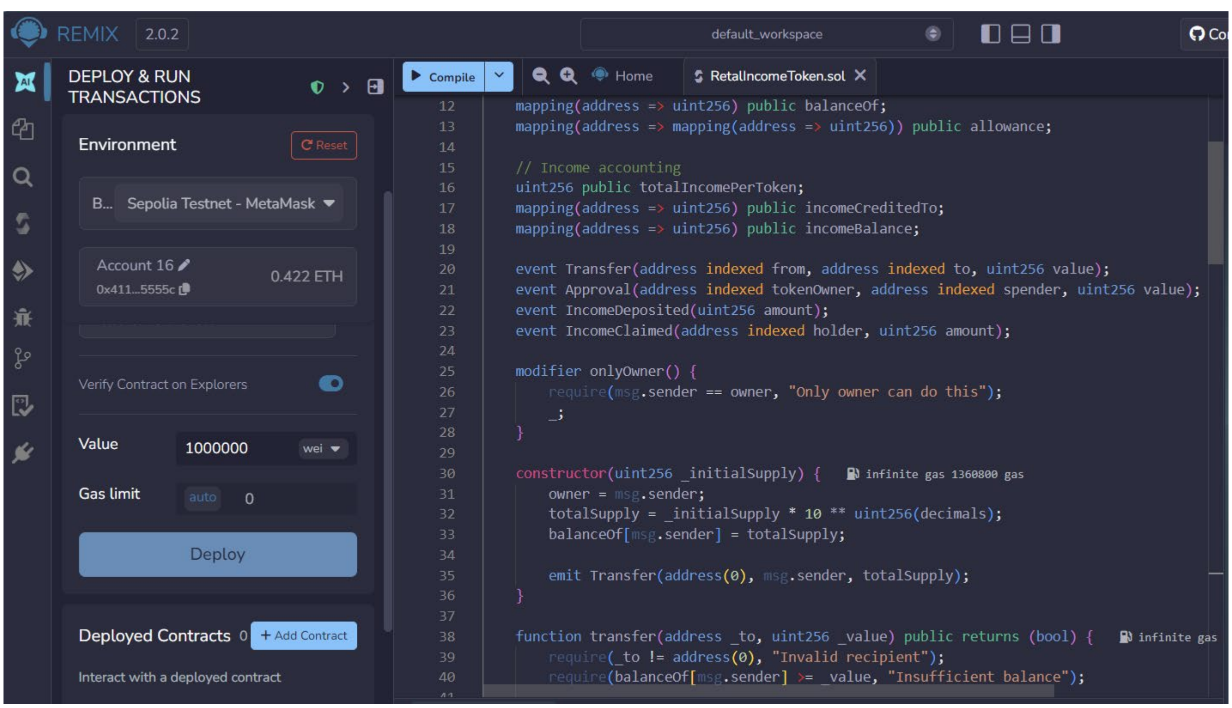
Implementation Evidence:



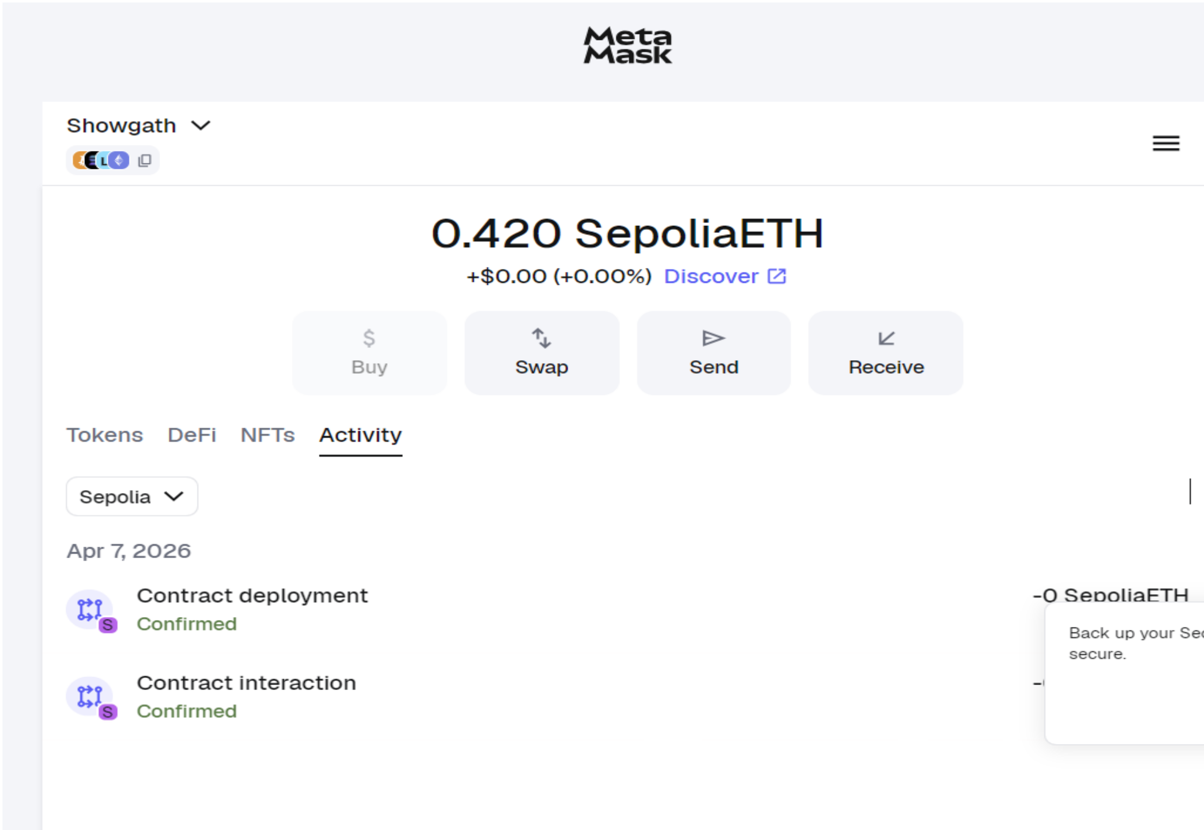
Screenshot 1



Screenshot 2



Screenshot 3



Screenshot 4

References

Arner, D.W., Barberis, J. and Buckley, R.P. (2017) 'FinTech and RegTech in a Nutshell', Journal of Banking Regulation, 19(4), pp. 1–14.

Catalini, C. and Gans, J.S. (2020) 'Some Simple Economics of the Blockchain', Communications of the ACM, 63(7), pp. 80–90.

Geltner, D., Miller, N.G., Clayton, J. and Eichholtz, P. (2013) Commercial Real Estate Analysis and Investments. 3rd edn. Mason, OH: South-Western Cengage Learning.

McMahan, J. (2006) Real Estate Investment and Management. New York: McGraw-Hill.

Nakamoto, S. (2008) Bitcoin: A Peer-to-Peer Electronic Cash System.

Szabo, N. (1997) Formalizing and Securing Relationships on Public Networks, First Monday, 2(9).

Tapscott, D. and Tapscott, A. (2016) Blockchain Revolution: How the Technology Behind Bitcoin and Other Cryptocurrencies Is Changing the World. New York: Penguin.